AI Campus Route Navigator

Assignment X_03 — Brief Report

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1. Objective

The project converts the supplied CU Technology Campus map into a weighted graph and compares PATHFINDER (Greedy Best-First Search) with ORBIT (A* Search). The comparison records route, path cost, explored nodes, and execution time.

2. Graph and file handling

Only locations visible on the supplied map are used. Connections and approximate walking costs are stored in campus.json rather than embedded in the search algorithms. This keeps the graph data separate from the routing logic.

3. Heuristic

The common heuristic is a scaled Euclidean estimate from the approximate map coordinates. Both agents use exactly the same heuristic so their search strategies can be compared fairly.

4. Algorithms

PATHFINDER: prioritizes h(n), so it mainly follows locations that appear close to the destination.
ORBIT: prioritizes g(n)+h(n), combining the cost already travelled with the estimated remaining cost.

5. CSE routing constraint

CSE Laboratory, CSE_AKC Seminar Hall, and CSE_Reflexon Room form the CSE zone. Entry is enforced through Tower 2 Front/Rear Entry → Lift Area → CSE location. While inside the zone, only CSE-labelled locations or the Lift Area may be visited; exiting requires Lift Area → Tower 2 Front/Rear Entry before returning to other campus locations.

6. Experimental results

Source → Destination	Agent	Cost (m)	Nodes	Time (s)
Reception of Calcutta University → Library	PATHFINDER	475.0	5	3.4e-05
Reception of Calcutta University → Library	ORBIT	475.0	10	7.3e-05
Canteen, CU Technology Campus → New Building 2 (Workshop Building)	PATHFINDER	290.0	3	8e-06
Canteen, CU Technology Campus → New Building 2 (Workshop Building)	ORBIT	290.0	3	6e-06
Entry Gate 1 (G1) → CSE Laboratory	PATHFINDER	685.0	14	3.1e-05
Entry Gate 1 (G1) → CSE Laboratory	ORBIT	685.0	40	6.6e-05
Auditorium Hall → CSE_AKC Seminar Hall	PATHFINDER	485.0	9	0.00014
Auditorium Hall → CSE_AKC Seminar Hall	ORBIT	485.0	33	5.7e-05
Library → Canteen, CU Technology Campus	PATHFINDER	440.0	19	3.5e-05
Library → Canteen, CU Technology Campus	ORBIT	440.0	10	2.1e-05
CSE_Reflexon Room → Entry Gate 4 (G4)	PATHFINDER	675.0	8	1.5e-05
CSE_Reflexon Room → Entry Gate 4 (G4)	ORBIT	480.0	7	1.1e-05

7. Observations and conclusion

The results illustrate that a node that appears closest to the destination is not necessarily part of the lowest-cost complete route. Greedy Best-First can explore fewer nodes but may choose a more expensive path. A* accounts for accumulated cost and can therefore prefer a route with a less attractive immediate heuristic. The CSE restriction can force additional detours that both algorithms must respect.

Execution times are machine-dependent. The graph connectivity and weights are modelling choices based on the supplied map and should be checked against the student's own campus knowledge before submission.